

Precision Mill — Bearing Failure & Replacement Manual

Failure signature

Progressive bearing wear commonly appears as increasing spindle vibration followed by a thermal rise. A simultaneous increase in vibration and temperature is a high-confidence indicator of bearing degradation.

Diagnostic thresholds

Normal vibration is approximately 1.5–2.8 mm/s. Sustained vibration above 4.0 mm/s requires inspection. Vibration above 6.0 mm/s with a thermal rise should be treated as critical.

Technician procedure

1) Stop the machine under the approved lockout/tagout procedure. 2) Inspect bearing housing and lubrication. 3) Check spindle alignment. 4) Replace the bearing if raceway damage or excessive play is confirmed. 5) Verify vibration after restart.

Replacement part

For the HX precision spindle family, the approved replacement is SKF-6205-2RS. Standard lead time is 12 days unless local inventory is available.

Historical pattern

Previous Machine_03 failures showed a rising vibration trend followed by temperature escalation. Repair logs attributed the failures to insufficient lubrication and raceway fatigue.